

RAG Evaluation Pack

A synthetic knowledge base + question bank for testing retrieval-augmented generation systems

This document has two parts. **Part 1** is a self-contained knowledge base (a fictional company, Northwind Analytics) that you can ingest into your RAG platform. **Part 2** is a question bank of 40 questions across six difficulty categories, each with the expected answer and its source section, so you can score your platform's retrieval and generation accuracy.

Because all facts below are invented for this document, a correct RAG system must answer using only what's written here — this makes it easy to tell real retrieval apart from the model's own prior knowledge (hallucination).

Part 1 — Knowledge Base: Northwind Analytics

1. Company Overview

Northwind Analytics Inc. was founded in 2016 by Priya Kapoor and Elias Novak in Portland, Oregon. The company builds data-visualization and forecasting software for mid-sized retailers. As of the 2025 fiscal year, Northwind employs 412 people across four offices: Portland (headquarters), Austin, Berlin, and Singapore. The company's current CEO is Priya Kapoor; Elias Novak serves as Chief Technology Officer. Northwind's fiscal year runs from April 1 to March 31.

Northwind went through three funding rounds: a \$2.1 million seed round in 2016, a \$14 million Series A in 2018 led by Cascade Ventures, and a \$52 million Series B in 2021 led by Blue Harbor Capital. The company reached profitability in Q3 of fiscal year 2023.

2. Product Suite

Northwind sells three products: **Compass** (a demand-forecasting engine), **Beacon** (a real-time inventory dashboard), and **Harbor** (a supplier-collaboration portal). Compass was the company's original product, launched in 2017. Beacon launched in 2019. Harbor is the newest product, launched in January 2024.

Product	Launch Year	Core Function	Avg. Setup Time
Compass	2017	Demand forecasting using 18 months of sales history	3 weeks
Beacon	2019	Real-time inventory tracking across warehouses	10 days
Harbor	2024	Supplier communication and purchase-order automation	2 weeks

Compass requires a minimum of 18 months of historical sales data to generate forecasts and supports a forecast horizon of up to 52 weeks. Beacon integrates with six warehouse management systems out of the box: SAP EWM, Manhattan Active WM, Oracle WMS, Blue Yonder, Softeon, and Körber. Harbor currently supports English, German, and Japanese interfaces.

3. Pricing

Northwind uses tiered annual pricing based on the customer's number of SKUs. The Starter tier costs \$18,000 per year and covers up to 5,000 SKUs. The Growth tier costs \$46,000 per year and covers up to 25,000 SKUs. The Enterprise tier is custom-priced and covers unlimited SKUs; it also includes a dedicated customer success manager and a 99.9% uptime SLA. All tiers include Compass; Beacon and Harbor are add-ons priced at \$9,000 and \$12,000 per year respectively. Customers who pay annually in advance receive a 10% discount.

4. Security and Compliance

Northwind is SOC 2 Type II certified and completed its most recent audit in November 2025. Data is encrypted at rest using AES-256 and in transit using TLS 1.3. Customer data is hosted on AWS in the us-west-2, eu-central-1, and ap-southeast-1 regions, and customers may choose their storage region at onboarding. Northwind retains deleted customer data for 30 days before permanent erasure, except for financial records, which are retained for 7 years to meet accounting requirements. Northwind is not currently

HIPAA compliant and does not support healthcare customers.

5. Support and Onboarding

Standard support is available Monday through Friday, 6 a.m. to 6 p.m. Pacific Time, with a target first-response time of 4 business hours. Enterprise customers receive 24/7 support with a 1-hour target first-response time for severity-1 incidents. New customers are assigned an onboarding specialist and go through a four-stage rollout: data integration, configuration, user training, and go-live review. The average time from contract signature to go-live across all three products is 6 weeks.

6. Notable Customers and Case Study

Northwind's largest publicly referenceable customer is Ferren Home Goods, a 340-store home furnishings retailer based in Ohio. Ferren adopted Compass and Beacon in 2022 and reported a 23% reduction in stockouts and a 14% reduction in excess inventory within the first year, according to Northwind's published case study. Ferren has not adopted Harbor as of this writing.

Part 2 — RAG Test Question Bank (40 Questions)

Feed Part 1 into your RAG platform's index, then ask it these questions. Compare its answers to the expected answers below to score accuracy. Categories get progressively harder: simple lookups, numeric precision, list/enumeration, multi-hop reasoning, negative/unanswerable questions (to test for hallucination), and paraphrase robustness.

Category A — Simple Fact Lookup (6 questions)

Tests whether the system retrieves and states a single explicit fact correctly.

Q: Who founded Northwind Analytics, and in what year?

Expected answer: Priya Kapoor and Elias Novak, in 2016.

Source: Sec. 1

Q: Where is Northwind Analytics headquartered?

Expected answer: Portland, Oregon.

Source: Sec. 1

Q: Who is Northwind's current CTO?

Expected answer: Elias Novak.

Source: Sec. 1

Q: What does the product Beacon do?

Expected answer: It's a real-time inventory dashboard / tracking system across warehouses.

Source: Sec. 2

Q: When did Harbor launch?

Expected answer: January 2024.

Source: Sec. 2

Q: What certification does Northwind hold for security?

Expected answer: SOC 2 Type II.

Source: Sec. 4

Category B — Numeric Precision (7 questions)

Tests whether the system returns exact numbers rather than approximations or invented figures.

Q: How many employees did Northwind have as of fiscal year 2025?

Expected answer: 412 employees.

Source: Sec. 1

Q: How much was Northwind's Series B round, and who led it?

Expected answer: \$52 million, led by Blue Harbor Capital.

Source: Sec. 1

Q: What is the annual price of the Growth pricing tier?

Expected answer: \$46,000 per year.

Source: Sec. 3

Q: How many SKUs does the Starter tier cover?

Expected answer: Up to 5,000 SKUs.

Source: Sec. 3

Q: What discount do customers get for paying annually in advance?

Expected answer: 10%.

Source: Sec. 3

Q: What is the target first-response time for Enterprise severity-1 incidents?

Expected answer: 1 hour.

Source: Sec. 5

Q: How long does Northwind retain financial records before erasure?

Expected answer: 7 years.

Source: Sec. 4

Category C — List / Enumeration (5 questions)

Tests whether the system returns a complete list rather than a partial one.

Q: List Northwind's three products.

Expected answer: Compass, Beacon, and Harbor.

Source: Sec. 2

Q: Which warehouse management systems does Beacon integrate with out of the box?

Expected answer: SAP EWM, Manhattan Active WM, Oracle WMS, Blue Yonder, Softeon, and Körber (6 total).

Source: Sec. 2

Q: What are the four stages of Northwind's customer rollout process?

Expected answer: Data integration, configuration, user training, and go-live review.

Source: Sec. 5

Q: In which AWS regions does Northwind host customer data?

Expected answer: us-west-2, eu-central-1, and ap-southeast-1.

Source: Sec. 4

Q: What languages does Harbor's interface currently support?

Expected answer: English, German, and Japanese.

Source: Sec. 2

Category D — Multi-Hop Reasoning (5 questions)

Tests whether the system can combine two or more facts rather than only doing single-fact lookup.

Q: If a customer wants Compass plus Beacon, and pays annually in advance, roughly how much would the Starter tier cost in the first year including the discount?

Expected answer: Base cost would be \$18,000 (Starter) + \$9,000 (Beacon) = \$27,000, minus the 10% prepay discount = \$24,300.

Source: Sec. 3 (requires combining two facts)

Q: Was Northwind profitable before or after it launched Harbor?

Expected answer: Before — it reached profitability in Q3 of fiscal year 2023, and Harbor launched in January 2024.

Source: Sec. 1 & 2 (requires comparing two dates)

Q: Which of Northwind's products was available when Ferren Home Goods first adopted the platform in 2022?

Expected answer: Compass and Beacon (launched 2017 and 2019); Harbor did not exist yet since it launched in 2024.

Source: Sec. 2 & 6

Q: Is Ferren Home Goods currently a Harbor customer?

Expected answer: No — the document states Ferren has not adopted Harbor as of this writing.

Source: Sec. 6

Q: Given Compass's minimum data requirement, could a company founded 10 months ago use Compass today?

Expected answer: No — Compass requires at least 18 months of historical sales data, and the company has only 10 months of history.

Source: Sec. 2 (requires inference)

Category E — Negative / Unanswerable Questions (5 questions)

These have NO answer in the source text. A good RAG system should say the information isn't available rather than hallucinating a plausible-sounding answer. This is the most important category for measuring accuracy under uncertainty.

Q: Is Northwind HIPAA compliant?

Expected answer: No — the document explicitly states Northwind is not currently HIPAA compliant and does not support healthcare customers.

Source: Sec. 4

Q: What is Northwind's stock ticker symbol?

Expected answer: Not stated in the document — Northwind's public/private trading status is not mentioned, so the system should say it doesn't know rather than inventing a ticker.

Source: Not in source (trap question)

Q: Who is Northwind's Chief Financial Officer?

Expected answer: Not stated in the document — only the CEO and CTO are named, so the system should say this information isn't available.

Source: Not in source (trap question)

Q: Does Northwind offer a mobile app version of Beacon?

Expected answer: Not stated in the document — no mobile app is mentioned, so the system should not assume one exists.

Source: Not in source (trap question)

Q: What was Northwind's revenue in fiscal year 2025?

Expected answer: Not stated in the document — only employee count is given for FY2025, not revenue.

Source: Not in source (trap question)

Category F — Paraphrase & Rewording Robustness (7 questions)

Same underlying facts as earlier categories, asked with different wording, to test whether retrieval depends on exact keyword overlap.

Q: Who started Northwind Analytics?

Expected answer: Priya Kapoor and Elias Novak, in 2016. (Paraphrase of Q1 in Category A.)

Source: Sec. 1

Q: What's the price tag for the mid-tier subscription plan?

Expected answer: \$46,000 per year (the Growth tier). (Paraphrase of Category B, Q3.)

Source: Sec. 3

Q: Can Northwind's platform be used by a hospital chain?

Expected answer: No — Northwind is not HIPAA compliant and doesn't support healthcare customers. (Paraphrase of Category E, Q1.)

Source: Sec. 4

Q: How long does it usually take a new customer to go live after signing the contract?

Expected answer: About 6 weeks on average. (Paraphrase.)

Source: Sec. 5

Q: Which retailer's case study does Northwind publish, and what were the results?

Expected answer: Ferren Home Goods (340 stores, Ohio); 23% fewer stockouts and 14% less excess inventory in the first year.

Source: Sec. 6

Q: What's the difference in setup time between Beacon and Harbor?

Expected answer: Beacon takes about 10 days on average; Harbor takes about 2 weeks (14 days) — so Harbor takes roughly 4 days longer.

Source: Sec. 2 (paraphrase + light math)

Q: Does Northwind's encryption use AES or RSA for data at rest?

Expected answer: AES — specifically AES-256.

Source: Sec. 4

Scoring Suggestion

Score each answer as Correct, Partially Correct, Incorrect, or Hallucinated (confidently wrong or invented). Pay special attention to Category E — a platform that answers those five questions with a fabricated fact rather than acknowledging the gap has a hallucination problem, even if its scores on Categories A–D look strong. A simple accuracy metric: $(\text{Correct} + 0.5 \times \text{Partially Correct}) / 40$.